Proposition Based Theory Specification (PBTS):
Guideline for the application in specification teams

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1. Introduction

Scientific theories aim to describe systematic regularities in the world. Often, they concern (causal) relations between concepts. Many theories aim to capture, for example, how concept A influences concept B, concept B influences concept C AND D etc.

There are multiple ways to describe theories either verbally (e.g., as a narrative or in a more structured form) or mathematically (e.g., in the form of equations or computational models). Scientific theories can according to classic theory of science approaches often be (re)described as a set of general implications (if - then statements) that describe the relation between propositions (i.e., statements that could be true or false). In many social sciences including psychology, however, most often, theories are presented as verbal narrative descriptions. The current theory crisis of psychology results from the fact that many of these verbal narrative descriptions of theories remain underspecified. That is, the description of the theory is vague so that predictions of the theory cannot be unambiguously and objectively derived. If it is not clear what a theory predicts, experiments cannot be used to update our beliefs about whether the theory is right or wrong. As a result, the value of any experiment is dramatically reduced, if theories are underspecified. Therefore, specification of a theory is crucial for an efficient development of knowledge.

The proposition based theory specification method (PBTS) provides a simple solution to these problems by offering a way

a) to analyze a theory, estimating how objectively a theory’s predictions can be derived from its verbal narrative description (i.e. theory reproducibility),
b) to explicate the implications and propositions contained in a theory in a formal and structured way, and
c) to advance the theory by solving problems, and filling gaps in a consensus process.

2. The Proposition Based Theory Specification (PBTS) approach

Theory description

To allow for testable predictions, a description of a theory should include

a) all relevant propositions and the implications that describe their relations,

b) definitions of all measurable and not directly measurable concepts that are included in the propositions and

c) (examples for) valid operationalizations and/or measurement methods of these concepts, including criteria for deciding when the proposition is considered to be true or false.

We refer to *propositions* as statements that can be true or false.

We refer to *implications* as statements that relate propositions in the form of IF- THEN statements. Implications are also statements that can be true or false. They, therefore, are a more complex form of propositions.

Propositions contain statements concerning *concepts*. Concepts can be not directly measurable (latent) variables (often referred to as constructs: e.g., intelligence, feeling of frustration, etc.) and directly measurable variables (e.g., behavior or properties of the person or situation: aggressive behavior, height of the person, temperature in the room, etc.). Concepts can therefore refer to constructs, behaviors, situations, and also other types of variables. We use the term concept broadly to also refer to the events and behaviours in the real world subsumed under them.

The simplest form of an implication is the following:

IF proposition A [referred to as *antecedence*] THEN proposition B [referred to as *consequence*]

An example from psychology is:

IF (a person feels frustrated[=true]) THEN (he/she shows aggressive behavior[=true])

This means that for all possible situations and all individuals, if frustration is true, then "shows aggressive behavior" is also true, or stated differently, frustration always leads to aggression. The statement IF (proposition A) is a short form for writing IF (proposition A=true); the part in squared brackets above [=true] can be omitted in such cases, since it is the default.

In psychological theories, propositions typically contain states of unobservable concepts (i.e. latent constructs), sometimes also observable concepts. For latent constructs, operationalizations (and/or measurement methods) have to be provided to decide whether a statement is true or false. For observable concepts, the conditions under which a proposition is considered true or false should be specified.

Sometimes, it is helpful to provide operationalizations on a population level (e.g., more than half of the subjects chose option A over option B) to allow for errors and statistical testing.

Typically, it is aimed that definitions and operationalizations are in line with common definitions and operationalizations used in the literature and validated before.

Note that often, issues of measurement and operationalization of concepts are not considered part of a theory. PBTS explicitly makes their specification part of the specification of a theory. This should foster the testability of theory, replicability, and improve efficient theory development by minimizing the underdetermination problem identified in the literature (e.g., Duhem-Quine thesis, Lakatos, 1970; Meehl, 1990).

For example, to make the implication introduced above testable, we (a) need to define how to measure the concept "a person feels frustrated" (and potentially define a threshold value from which we consider this to be true, in case a continuous measure is used). Furthermore, we (b) have to specify how to measure the observable variable "aggressive behavior" and how to decide whether a behavior is considered "aggressive" or not. Spelling out the example already leads to interesting insights. For example, we learn that the current specification of the theory implies binary states of frustration (yes vs. no) and no gradual increase in the likelihood of aggressive behavior with increasing frustration. Furthermore, it becomes clear that for a theory to make clear-cut predictions, we need to invest considerable thought in operationalizations and measurement of the relevant concepts.

Details of implication statements

In order to capture complex and computational theories as well as relations between continuous concepts, we allow the antecedence and consequence to also include multiple propositions that can be combined by any logical operators (e.g., AND, OR, NOT), qualified by probabilistic statements, mathematical functions, or user-defined uncommon operators.

For example:

If (concept A) *AND* (concept B) THEN (concept C)

If (concept A) *AND NOT* (concept B) THEN (concept C) *OR* (concept D)

If (concept A) THEN (concept B) = (concept C) * (concept D)^(concept E)

If (concept A) THEN (*5% of the population shows* concept B)

If ((concept A) and (concept B) *are balanced*) THEN (concept C)
[comment: it has to be explicitly specified when the user-defined uncommon operator (e.g. "*are balanced"*) produces true or false values in step 3 below)]

Auxiliary assumptions

Moreover, theories can include further auxiliary assumptions that are needed to make a theory testable or applicable. These can include pre-conditions under which the theory is applicable or can be tested that are not an inherent part of the theoretical assumptions (i.e. propositions) and operationalizations/definitions.

3. Procedure for Deriving Theory Reproducibility Indicators and PBTS-Based Theory Specifications

To apply PBTS, each team completes the following eight steps. Steps 2–7a are primarily designed to derive all necessary theory reproducibility indicators. To this end, please ensure that you stay as close as possible to the original narrative description of the theory (except for Step 3b). Steps 7b and 8 focus on generating a final and potentially advanced specification of the theory. In these steps, you may contribute your own knowledge to the specification, or add information from other sources; however, be sure to clearly mark these instances as described below. Please complete all tables in full.

General steps proposed by PBTS

Preparation (within the whole group)
(Step 1) Select the theory and the version of the theory to be specified, and agree on these sources for the theory specification with fellow raters.

Independent Theory Specification (each rater by themselves)
(Step 2) State a set of implications (i.e., IF- THEN statements) and all included propositions that together fully describe the theory.
(Step 3) Define all concepts and uncommon operators (if used), and operationalize all propositions, listed under Step 2.
(Step 4) Specify auxiliary assumptions of the theory.
(Step 5) Double-check for problems in the theory specification and note important insights.
(Step 6) Rate various aspects of the theory.

Evaluate Rater Agreement (in dyads)
(Step 7a) Compare individual specifications and evaluate rater agreement.

Generate Consensus Theory Specification (within the whole group)
(Step 7b) Achieve consensus on a final version.
(Step 8) Analyze theory specification issues.

More detailed instructions on how to complete Steps 1 to 8 are provided below, please read these guidelines first before starting the theory specification.

4. Detailed Explanations and Instructions for Steps 1 to 8

(Step 1; whole group) Select the theory and the version of the theory to be specified, and agree on these sources for the theory specification with fellow raters.

(a) Decide which theory, and which version of the theory (e.g., original, revised, or latest) will be specified, including the name of the theory, the author, and the original reference.

(b) Select 2–3 key sources for the theory specification (e.g., the original article, relevant journal papers, or textbook chapters).

(c) If the original theory is highly extensive (e.g., published as a book), focus on its core content as represented in typical journal articles or summary chapters. Additional aspects from larger works may be addressed in future research. Any deviations from using the original source must be explicitly documented in Table 1.

(d) For each selected source, specify the relevant sections (e.g., "p. 1, line 1 – p. 2, line 20") that include the theory's key causal relations, concept definitions, operationalizations, and auxiliary assumptions.

(e) Before beginning the specification, all raters should agree on the selected sources to ensure consistency and efficiency. In cases of uncertainty, include more text rather than less. The final version of Table 1 should include all agreed sources with page/line information, DOI, and, for books, the edition used.

Table 1: THEORY AND SOURCE SELECTION (specified in Step 1)
= version of the theory and text sections used for the following theory specification

Please state the theory and version of the theory you want to specify. Cite the sources used, indicating the pages and line details of the relevant sections for the following theory specification or link to the shared document in which the text sections are contained. The raters should have agreed on the sources before conducting further steps. However, sometimes necessary updates to the list might become clear only after conducting further steps.

Theory and Version of the Theory to be specified:		
Nr	Source	Pages and Lines used
1		
2		
3		

Explanation for Source Selection (if necessary):	

(Step 2; individually) State a set of implications (i.e., IF- THEN statements) and all included propositions that together fully describe the theory.

(a) Propositions are statements concerning measurable and non-measurable concepts that can be true or false, and represent an implication's antecedence and consequence.

(b) Propositions can be combined by any logical or mathematical operators, qualifiers, or, if necessary, user-defined uncommon operators (that have to be defined and operationalized).

(c) It is often possible to make certain conditions of a theory either part of the propositions or the definition/operationalizations. To keep the propositions simple, these conditions should be part of the definition/operationalizations if they do not belong to the core causal relations of the theory.

(d) In this step, remain as close as possible to the original verbal narrative while interpreting the text in accordance with the authors' intended meaning.

(e) In the column "DER-Rating," indicate the degree to which each implication corresponds to the original text.

(f) In the "Reference" column, include the text snippet and citation from which each implication was obtained.

(g) Specific predictions concerning functional relations between concepts can be put in the "THEN" part of the implication. For example: "IF" (A takes values in the Range [0-1]) "THEN" (B increases with increasing A)). A similar logic can be applied for more complex functional forms by changing the "THEN" component as follows: "THEN" (A = f(B)); with f() indicating a monotonic/linear/exponentially increasing function (as specified in the Table for uncommon functions in step 3); or "THEN" (A = B^alpha); with alpha > 0.

(h) Functions can have different degrees of precision. It is fine to start with relatively vague functions (e.g., f() … monotonic increasing function vs. linearly increasing function).

Refresher on logical operators:

- AND (Conjunction): True only if all connected propositions are true
    - IF A **AND** B THEN C → Both A and B must be true for C to occur
- OR (Disjunction): True if at least one connected proposition is true
    - IF A **OR** B THEN C → Either A or B or Both must be true for C to occur
- NOT (Negation): True when the connected proposition is false, and vice versa
- IF A AND NOT B THEN C →A must be true and B must be false for C to occur

(i) Below the table, specify which of the listed implications are core to the theory, meaning that falsifying them would amount to a falsification of the theory itself, and which implications are less central, where falsification would instead indicate a need for theoretical revision or refinement.

<u>Pragmatic recommendations</u>

- Use parentheses for a proposition and italicize uncommon operators.

Table 2: IMPLICATION (Specified in Step 2)

= Main statements that describe the theory.

Please formalize the implications using IF ... THEN ... statements, where IF = antecedence, THEN = consequence. Use also OR, AND or any logical or mathematical operators if you need it. In case you use uncommon operators, please also explain the operators briefly (in Table 3).

In the column "DER-Rating," please rate how closely each implication reflects the original text by entering the following number:

4 - directly taken from the text
3 - very close to the text
2 - mentioned or partially reflected in the text
1 - weakly implied by the text
0 - not mentioned in the text

In the column "Reference," please provide the specific text snippet or passage that supports each implication (including the reference).

Example "Modified Facial Feedback Hypothesis" (Noah et al., 2018):

(Antecedence:) IF (the facial muscles related to emotional expressions *are activated*) AND NOT (feels observed)

(Consequence:) THEN (positive emotions *are induced*) AND (stimuli *are evaluated more positively*).

Nr	IF	THEN	Reference	Comments	DER-Rating
1					
2					

The Core Implications of the Theory are (just provide the numbers, e.g., implications #1, #3, and #7):

(Step 3; individually) Define all concepts and uncommon operators (if used), and operationalize all propositions listed under Step 2.

Step 3a: Extract Information as provided by the author

(a) In Table 3.1: Definitions

  (i) Extract all concepts and uncommon operators from Table 2 and copy each of them in a separate row in Table 3.1.

  (ii) Add definitions for each concept/uncommon operator as provided (or referred to) by the author of the theory.

    (1) Common operators do not need to be defined. Common operators are links or functions that occur as standard in logic or mathematics (e.g., AND, OR, NOT, *, +, -, /, ^)

  (iii) In the column "DER-Rating," indicate the degree to which each definition corresponds to the original text. If the original text does not contain a corresponding definition, indicate a 0.

  (iv) In the "Reference" column, include the text snippet and citation from which each definition was obtained.

(b) In Table 3.2: Operationalizations

  (i) Extract all propositions from Table 2 and copy each of them in a separate row in Table 3.2.

  (ii) Add operationalizations for each proposition as provided (or referred to) by the author of the theory.

    (1) In cases in which propositions contain multiple concepts that need to be measured separately to determine the truth status of a proposition, operationalizations for each concept should be provided (e.g., "*increases with* probability * value" - in this case operationalizations for probability and value are needed and should be provided in separate lines but in the same row of the table).

  (iii) In the column "DER-Rating," indicate the degree to which each operationalization corresponds to the original text. If the original text does not contain a corresponding operationalization, indicate a 0.

  (iv) In the "Reference" column, include the text snippet and citation from which each operationalization was obtained.

Step 3b: Generate preliminary suggestions for missing information

It can be the case that information concerning definitions of concepts or operationalizations of propositions is not directly provided by the author of the theory. You have marked these cases in the tables 3.1 and 3.2 with a DER-Rating of 0 and the respective content fields should be empty. In the later steps of theory specification and advancement you will develop solutions for these cases with your specification team. Please now generate first solutions for all missing information by imputing what could be reasonable or what the author could have had in mind. This should allow evaluating the theory better and provides a basis for later discussions in the team. Specifically, go through the definitions and operationalizations in Tables 3.1 and 3.2. respectively, and add information concerning reasonable definitions (e.g., from your knowledge or from the literature) and reasonable or ideas for additional operationalizations. **Importantly, please mark this imputed information clearly by putting a marker "DER" before this information.** An example could be completing a theory that concerns aggression but does not define it with the following common definition from a different source: "DER: Aggression is defined as: Any form of behavior directed toward the goal of harming or injuring another living being who is motivated to avoid such treatment - reference: Baron & Richardson, 1994)."

Step 3b can easily take overly much time. We recommend setting an explicit time limit for this step (e.g., max two hours).

Table 3.1: DEFINITIONS OF CONCEPTS / UNCOMMON OPERATORS (specified in Step 3)

= further specification of concepts / uncommon operators used within the formulation of the implications in Step 2.

Please copy all concepts / uncommon operators you have used in your implications in Step 2 into Table 3.1. Add all definitions of these concepts / uncommon operators that you can find in the texts.

In the column "DER-Rating," please rate how closely each definition reflects the original text by entering the following number:

- 4 - directly taken from the text
- 3 - very close to the text
- 2 - mentioned or partially reflected in the text
- 1 - weakly implied by the text
- 0 - not mentioned in the text

In the column "Reference," please provide the specific text snippet or passage that supports each definition.

Example "Modified Facial Feedback Hypothesis"
Note, that 7 concepts and operators are contained in the example, only the first three are specified in the following.

Nr	Concept / Uncommon Operator	Verbal definition	Reference	Comments	DER-Rating
1	facial muscles related to emotional expressions	Facial muscles related to emotional expressions	"zygomaticus major muscle that is used in smiling [...] the orbicularis oris muscle which inhibits muscle activity associated with smiling" Noah et al., (2018), p. 658		4

2	[the facial muscles related to emotional expressions] *are activated*	Activation of facial muscles expressions	"holding a pen either between their teeth or between their lips" Noah et al., (2018), p. 657		4
3	feels observed	A person feels that (s)he is watched by others	"influence of feeling observed on people's reliance on their inner cues" Noah et al., (2018), p. 657		4

Nr	Concept/Uncommon Operator	Verbal Definition	Reference	Comments	DER-Rating
1					
2					
3					

Table 3.2: OPERATIONALIZATIONS OF PROPOSITIONS (specified in Step 3)

= further specification of propositions used within the formulation of the implications in Step 2.

Please copy all propositions you have used in your implications in Step 2 into Table 3.2. Add all operationalization of these propositions that you can find in the texts.

In the column "DER-Rating," please rate how closely each operationalization reflects the original text by entering the following number:

4 - directly taken from the text
3 - very close to the text
2 - mentioned or partially reflected in the text
1 - weakly implied by the text
0 - not mentioned in the text

In the column "Reference," please provide the specific text snippet or passage that supports each operationalization.

Example "Modified Facial Feedback Hypothesis"
Note, that only the first two propositions are operationalized in the following.

Nr	Proposition	Measurement/ Operationalization	Reference	Comments	DER-Rating

1	the facial muscles related to emotional expressions *are activated*	Manipulation: holding a pen between the teeth vs. the lips	“holding a pen either between their teeth or between their lips” Noah et al., (2018), p657	methods for measuring activation are not provided	4
2	feels observed	Manipulation: being aware that they are videotaped vs. not	“without a camera [...], and with a camera in front of the participants during the experiment…” Noah et al. (2018), p658	methods for measuring feelings not provided	4

Nr	Proposition	Measurement/ Operationalization	Reference	Comments	DER-Rating
1					
2					
3					
4					

(Step 4; individually) Specify auxiliary assumptions of the theory.

In this section, further assumptions and pre-conditions of the theory to hold can be listed.

(a) Auxiliary assumptions and pre-conditions that are mentioned by the author of the theory should be listed in a numbered list, one per line. They concern requirements that should be met for all implications.

(b) For each line, an explanation and/or text snipped including reference should be added in a separate column that describes the origin of the auxiliary assumption. Moreover, a brief definition of that assumption should be given.

An example of an auxiliary assumption could be: “theory is only applicable in Western countries”, with “Western countries” defined in the column next to it.

Table 4: AUXILIARY ASSUMPTIONS (specified in Step 4) **[only if necessary - potentially skip]** = assumptions that are secondary to the theory under investigation. They describe necessary pre-conditions to test a theory or boundary conditions. Only aspects that are not already covered in step 3 should be listed here. All auxiliary assumptions mentioned by the authors of the theory should be covered. In the column “DER-Rating,” please rate how closely each auxiliary assumption reflects the original text by entering the following number: 4 - directly taken from the text 3 - very close to the text 2 - mentioned or partially reflected in the text 1 - weakly implied by the text 0 - not mentioned in the text In the column “Reference,” please provide the specific text snippet or passage that supports each auxiliary assumption.
Example "Modified Facial Feedback Hypothesis": The authors did not specify auxiliary assumptions in the sources, therefore the table would be kept empty. Auxiliary assumption(s) could for example be: Implication 1 only holds, if the muscles were not activated before by another cause or Implication 1 only holds in Western countries.

Nr.	Description	Definition (Optional: Operationalization)	Reference	Comments	DER-Rating
1					
2					

(Step 5; individually) Identifying and Reporting Problems in the Theory Specification and Note Important Insights.

Formalizing a theory can reveal underspecified elements, tautologies, logical contradictions, or derived assumptions not stated in the original text. Identifying these issues helps refine and clarify the theory.

- Record problems in Table 5.1.
- Note insights, further directions, and open questions in Table 5.2.

Common Issues to Check
1.) A **tautology ([TAU])** occurs when the antecedent (IF) and consequence (THEN) use overlapping or identical variables, often hidden by similar wording or measurement.
→ List each in Table 5.1 and mark [TAU].

2.) A **contradiction ([CON])** occurs when two implications logically exclude each other (e.g., "IF frustration THEN aggression" vs. "IF frustration THEN NOT aggression").
→ List each in Table 5.1 and mark [CON].

3.) An **underspecification ([USPEC])** means relations between concepts or their definitions are too vague to derive clear propositions.
→ List each in Table 5.1 and mark [USPEC], adding a short explanation in another column.
→ Typical cases include:

- Conceptual clarity: Overlapping or poorly defined constructs.
- Timing issues: Unclear temporal sequence or dynamics of variables.

During the specification process, you may identify insights, further directions, and open questions, like:

- Important or unexpected aspects of the theory,
- Areas requiring further specification,
- Directions for future theory development or integration.

→ List each point in Table 5.2, one per row.

Table 5.1: Problems (specified in Step 5)
= Basic requirements for any scientific theory to hold are the absence of tautologies and logical contradictions. In addition, precise definitions of all aspects of the theory and their interrelation should be provided. Are there any tautologies in the specification of the theory, or is there a risk that a tautology might appear for some operationalizations of variables? Note each potential problem on a separate line and mark them with [TAU]. Are there any contradictions in the specification of the theory, or is there a risk that a contradiction might appear for some specifications? Note each potential problem on a separate line and mark them with [CON]. Are there issues in the verbal description of the theory that are underspecified to a degree that deriving implications and propositions is hard to impossible? Issues that are already solved by operationalizations and definitions are excluded here. List them, one per row, below and mark them with [USPEC]. Provide a short explanation in a separate column.

Nr.	Description	Markers	Reference / Comments
1			
2			

Table 5.2: Insights, Further directions and Open Questions (Specified in Step 5):
In the process of theory specification, one often gains insights concerning i. particularly important and unexpected aspects of the theory, ii. aspects that would be important to specify further, and iii. direction for further theory development or theory integration. These insights should be listed below using one row each.

Nr.	Description	Markers	Reference / Comments
1			
2			

Continue with Step 6 only, if you are satisfied with your overall theory specification and feel that you have finished the independent part of the theory specification procedure.

(Step 6; individually) Rate various aspects of the theory.

Once you have finished specifying all components of the theory, please complete Table 6, which addresses several key aspects of the theory you have outlined. This will facilitate comparison across different theories and provide insight into how easily the theory could be specified, as well as how clearly it had already been specified in the existing literature.

Table 6: Theory Ratings	
Please provide overall ratings concerning the following aspects of the theory you specified in steps 2 to 5. These ratings should be based on the version of the theory and sources specified in step 1 and reflect your experiences from individual theory specification in steps 2 to 5. Aspects that you added to the theory or derived from other sources should not be considered in the ratings. Please indicate whether you agree or not to the following statements using the following 7-point scale: 1: Strongly Disagree 2: Disagree 3: Somewhat Disagree 4: Neither Agree nor Disagree / Neutral 5: Somewhat Agree 6: Agree 7: Strongly Agree	
6.1 Theory Specification	
1) The core mechanisms that the theory proposed are well described by the author.	
2) The propositions and implications that constitute the theory are clearly stated.	
3) The relations between all concepts are articulated in an unambiguous manner.	
4) Definitions for all relevant concepts are provided.	
5) Operationalizations for all relevant propositions are provided.	
6) The various aspects of the theory are overall very well specified by the author.	
7) The original formulation of the theory contains risks for tautologies to appear.	
8) The original formulation of the theory contains risks for inconsistencies.	
9) The original formulation of the theory contains risks for underspecification.	
6.2 Subjective Assessment of Theory Specification	
1) It was easy for me to extract the propositions and implications from the authors' descriptions of the theory.	

2) Extracting definitions for the relevant variables was easily possible.	
3) Extracting operationalizations for the relevant variables was easy.	
<u>6.3 Properties of the resulting specified versions of the theories</u>	
6.3.1 Complexity	
1) The theory contains many variables and relations compared to other theories in social psychology.	
2) The theory contains many complex relations between variables compared to other theories in social psychology.	
3) The theory is overall very complex compared to other theories in social psychology.	
4) How many implications are included in the theory? (lines in Table 2)	______
5) How many concepts/uncommon operators are contained in the specification? (lines in Table 3.1)	______
6) How many propositions are contained in the specification? (lines in Table 3.2)	_______
6.3.2 Empirical content of the theory generality (i.e. inclusiveness of the if-components / antecedent) and precision (i.e. the exclusiveness of the then-components / consequence).	
6.3.2.1 Generality	
1) The theory is very general in that it can be applied to many different situations and persons.	
2) There are boundary conditions explicated that reduce the generality of the theory.	
3) The antecedence of the theory (IF component) is broad and does not constrain the application area substantially.	
4) The auxiliary assumptions constrain the application area of the theory substantially.	
5) The operationalizations provided allow to test the theory broadly and in many situations.	
6.3.2.2 Precision of the theory	
1) The theory is very precise in its predictions.	
2) The THEN component of the theory specifies clearly what should happen.	

3) The THEN component of the theory is restrictive in that it forbids many alternative consequences.	
4) The theory makes testable predictions on multiple dependent variables.	
Uniqueness of theory predictions	
1) The theory makes many predictions that differ from predictions of other theories I know of.	

(Step 7a; in dyads) Compare individual specifications and evaluate rater agreement

In this step, raters should work in dyads. If the group consists of an uneven number of raters, one rater will have to skip Step 7a.

The purpose of this assessment is to systematically capture the content-based agreement of (7.1) Implications and (7.2) Definitions/Operationalizations formulated by two raters.

(7.1) Implications

1) **Listing implications**

   a) Each implication created by Rater 1 (i.e. each line from their Table 2) is entered in a separate row of Table 7.1 below. Rater 1 enters their implications under "Version Rater 1".

   b) Please make sure to also copy any markers included (for example [DER] for derived aspects of the theory), as these markers will be important for the ratings in this step.

2) **Matching implications**

   a) Rater 1 tries to match the implications by Rater 2 as well as possible by adding Rater 2's implications under "Version Rater 2". Implications that address the same aspect (might be more than one implication) should be in the same line even if they are not fully identical.

   b) If some aspect is only captured by one of the raters, the corresponding cell for the other rater remains empty.

   c) Rater 2 double-checks the matching. In case of disagreement, Rater 2 makes an alternative suggestion and Rater 1 and 2 find a common solution for the matching.

3) **Assessment of the Agreement:**

   a) We aim to get **two independent ratings of agreement** of the implications based on the matched list in Table 7.1 generated in the previous matching step. Therefore, Rater 1 and 2 should independently provide their agreement ratings in Table 7.1. For this, please generate two copies of Table 7.1 after finishing the matching process, and indicate your name on your copy before adding your ratings to your copy of the table.

   b) Each rater assesses each row for content-based agreement. It is recommended that ratings are not too strict so that minor differences are not attributed to the author of the original theory, especially if they reflect natural variation in expression. Linguistic differences in wording, such as synonyms, capitalization, and the use of definite or indefinite articles (e.g., "a person" vs. "the person"), should be disregarded. Furthermore, consider the following cases as complete agreement (with an explanation in the comments):

      i) If one rater expresses an implication as a single statement while the other splits it into two or more implications.

      ii) If a statement appears in one rater's implication but is embedded within the verbal definition/operationalization of the other.

   c) The following rating scale should be used:

- **4: Full Agreement** - Implications are entirely aligned in their content meaning, even if synonyms are used.
- **3: 75% Agreement** - Implications are largely aligned in content, with only minor differences.
- **2: 50% Agreement** - Implications show a moderate degree of alignment, with some significant differences.
- **1: 25% Agreement** - Implications show little alignment in content, with several notable differences.
- **0: No Agreement** - Implications show no content-based agreement.
- **88: Agreement on missing** - Both raters indicate this verbal implication to be missing in the text.
- **99: Disagreement on missing** - Only one of the two raters indicates this implication to be missing in the text.

d) Please document any considerations and uncertainties in the comment column.

Table 7.1 Evaluation of Rater Agreement on Implications (from Tables 2)

Please rate the agreement between the implications by entering the following number:

4 - Full Agreement
3 - 75% Agreement
2 - 50% Agreement
1 - 25% Agreement
0 - No Agreement
88 - Agreement on missing
99 - Disagreement on missing

Nr	Implication - Version Rater 1		Implication - Version Rater 2		Comment	Rating Rater 1
1	*Example:* *IF (the facial muscles related to emotional expressions are activated) AND NOT ((*feels observed*))*	*THEN (positive emotions are induced) AND (stimuli are evaluated more positively*) (1)	*IF (the facial muscles are activated) AND NOT (consciously monitored)*	*THEN (positive feelings are induced) AND (stimuli receive a more positive evaluation*) (1)	R1: Synonyms: "emotions" and "feelings"; "are evaluated more positively" and "receive a more positive evaluation" Differences: *(the facial muscles **related to emotional expressions)*** in V1 vs. not in V2; (**feels observed** vs. **consciously** monitored)	3

Nr	Implication - Version Rater 1		Implication - Version Rater 2		Comments	Rating Rater (1 / 2)
	IF	THEN	IF	THEN		(Name)
1						
2						
3						

(7.2) Definitions of Concepts and Operators; Operationalizations of Propositions

The same procedure as described in (7.1) should be applied to the definitions of concepts and uncommon operators (Table 7.2.1), as well as the Operationalization of Propositions (Table 7.2.2) including Listing, Sorting & Matching, Assessment and Discussion of the Agreement.

1) Assessment of Agreement - additional remarks
    a) Please note again that the evaluation in this step should not be overly strict.
        i) If Rater 1 has defined/operationalized a concept/proposition that Rater 2 has divided into two or more statements, these can be considered as matching.
        ii) Similarly, if Rater 1 has already addressed part of the concept in the propositions (Step 2), while Rater 2 only incorporates it in Step 3, this can also be considered a complete agreement.
    b) If both raters leave a field blank indicating agreement that a definition/operationalization is missing, record a rating of 88. Leaving the field blank indicates that neither rater could identify a definition/operationalization in the text.
    c) If one of the two raters has a blank field, indicating a disagreement on whether a definition / operationalization is missing, record a rating of 99. This indicates disagreement about whether a definition/operationalization was provided in the text.
    d) The following rating scale should be applied:
        - **4: Full Agreement** - Definition/operationalization are entirely aligned in their content meaning, even if synonyms are used.
        - **3: 75% Agreement** - Definition/operationalization are largely aligned in content, with only minor differences.
        - **2: 50% Agreement** - Definition/operationalization show a moderate degree of alignment, with some significant differences.
        - **1: 25% Agreement** - Definition/operationalization show little alignment in content, with several notable differences.
        - **0: No Agreement** - Definition/operationalization show no content-based agreement.

- **88: Agreement on missing -** Both raters indicate this verbal definition/operationalization to be missing in the text.
- **99: Disagreement on missing -** Only one of the two raters indicates this verbal definition/operationalization to be missing in the text.

Table 7.2.1: Evaluation of Rater Agreement on Definitions of Concepts and Operators (from Tables 3.1)

Please rate the agreement between the concepts/uncommon operators by entering the following number:

4 - Full Agreement
3 - 75% Agreement
2 - 50% Agreement
1 - 25% Agreement
0 - No Agreement
88 - Agreement on missing
99 - Disagreement on missing

Nr	Rater 1		Rater 2		Comments	Rating Rater (1 / 2)
0	Concept Label	Verbal Definition	Concept Label	Verbal Definition		(Name)
1						
2						

Table 7.2.1: Evaluation of Rater Agreement on Operationalizations of Propositions (from Tables 3.2)

Please rate the agreement between the operationalizations by entering the following number:

4 - Full Agreement
3 - 75% Agreement
2 - 50% Agreement
1 - 25% Agreement
0 - No Agreement
88 - Agreement on missing
99 - Disagreement on missing

Nr	Rater 1	Rater 2	Comments	Rating Rater (1 / 2)

0	Proposition	Operationalization	Proposition	Operationalization		(Name)
1						

(Step 7b; whole group) Achieve consensus on a final consensus version

(7.3.1) Final Consensus Version: Implications

1) Back within the whole group, all raters try to find a final consensus version of the Implications they can agree on and list it in Table 7.3.1 below.

2) Once you added an implication, please indicate a DER-Rating and provide a reference to where in the text this idea is expressed in the respective columns of the Table 7.3.1.

3) Theory Advancement: If the original authors did not provide certain implications but they are necessary to make the theory complete or to close loopholes, derive them from your previous knowledge or the scientific literature in psychology. Mark them with “[DER]” and provide the source or a brief explanation in the “Reference” column if possible.

4) Once the group has agreed on a final version, number all implications, propositions, concepts, operators, and operationalizations consistently. Use clear letter–number codes (e.g., P1, C2, O3) to maintain clarity and enable easy cross-referencing between tables.

<u>Numbering system:</u>

- **Implication**: IF (P[1]=true) – THEN (P[2]=true) statements
   I1, I2, I3,… I_N (the number is in the first column of Table 7.3.1)

- **Proposition**: statements that can be true or false
   P1, P2, P3 … P_N

- **Concept**: any construct, behavior, or variable
   C1, C2, C3 … C_N

- **Operator**: any qualifiers for concepts, equations, or operators that combine them
  O1, O2, O3…O_N

- **OperationaliZations**: any possibility to measure/operationalize whether a proposition is true or false, mainly related to propositions, sometimes also to concepts
   Z1_P1, Z2_P1, Z3_C1, Z4_O1 … Z[N]_P/C/O[M]

Examples:

- Typically, an implication consists of at least two propositions
  - e.g., I1: IF (P1: (C1: Frustration) THEN (P2: (C2: Aggression))
  - Each proposition consists of any combination of concepts C and operators O
    - e.g., (P3: (C3: Anger) (O1: *increases*)); or:
    - (P3: (C3: Anger) (O1: *increases with*) (C4: Individuals Arousal)).

Table 7.3.1: Final Consensus Implications = Main statements that describe the theory. Please formalize the implications using IF ... THEN ... statements, where IF = antecedence, THEN = consequence. Use also OR, AND or any logical or mathematical operators if you need it. In case you use uncommon operators, please also explain the operators briefly (in Table 7.3.2). In the column "DER-Rating," please rate how closely each implication reflects the original text by entering the following number: 4 - directly taken from the text 3 - very close to the text 2 - mentioned or partially reflected in the text 1 - weakly implied by the text 0 - not mentioned in the text In the column "Reference," please provide the specific text snippet or passage that supports each implication.					
Nr	IF	THEN	Reference	Comments	DER-Rating
1					
2					

(7.3.2) Final Consensus Version: Definitions and Operationalizations

The raters should now decide on a final version of all verbal definitions and operationalizations for the concepts used in Table 7.3.1.

1.) Theory Advancement:

   a) In cases where definitions are not provided by the author, derive them from your previous knowledge or the scientific literature in psychology. Mark them with "[DER]" and provide the source or a brief explanation in an extra column if possible. Each team member should contribute his / her ideas individually developed in Step 3b (above).

   b) In cases where operationalizations are not provided by the author, derive them from your prior knowledge or the scientific literature in psychology. Preferably, use operationalizations that were used by the author of the theory in other empirical work or other authors that claimed to test the theory empirically. Mark them with "[DER]" and provide the source in the "Reference" column or a brief explanation. Each team member should contribute his / her ideas individually developed in Step 3b (above).

i) Double-check whether these operationalizations have generated / can be expected to generate the typically found effects in the standard paradigms.

ii) It is preferable to have various operationalizations that can be applied in different situations since otherwise, the theory often cannot be applied, and the empirical content of the theory is reduced. If possible provide operationalizations for

(1) experimental situations (manipulation & manipulation check),

(2) surveys (e.g., questions, this could also be the same as before), and

(3) practical applications and behavior outside the lab (e.g., external ratings of the situation or the person that can be applied post-hoc or using observation).

Table 7.3.2.1: Final Consensus Definitions

= further specification of concepts and operators used within the implications.

Please copy all concepts and uncommon operators you have used in your implications in Table 7.3.1 into this table. Add all verbal definitions of these concepts/operators.

In the column "DER-Rating," please rate how closely each definition reflects the original text by entering the following number:

4 - directly taken from the text
3 - very close to the text
2 - mentioned or partially reflected in the text
1 - weakly implied by the text
0 - not mentioned in the text

In the column "Reference," please provide the specific text snippet or passage that supports each definition.

Nr	Concept / Operator	Verbal Definition	Reference	Comments	DER-Rating
1					
2					

Table 7.3.2.2: Final Consensus Operationalizations

= further specification of propositions used within the formulation of the implications.

Please copy all propositions you have used in your implications Table 7.3.1 into this table. Add all operationalization of these propositions.

In the column “DER-Rating,” please rate how closely each operationalization reflects the original text by entering the following number: 4 - directly taken from the text 3 - very close to the text 2 - mentioned or partially reflected in the text 1 - weakly implied by the text 0 - not mentioned in the text In the column “Reference,” please provide the specific text snippet or passage that supports each operationalization.					
Nr	Proposition	Measurement / Operationalization	Reference	Comments	DER-Rating
1					
2					

(7.3.3) Final Consensus Version: Auxiliary Assumptions

In this section, list all auxiliary assumptions and preconditions required for the theory to hold. Include both those explicitly mentioned by the author and additional ones you have identified yourself, marking the latter with [DER] for derived. Each assumption should be presented in a numbered list, one per line, with a short explanation or text snippet (including reference) that indicates its source and a brief definition of the assumption.

Example: The theory only holds if: (individual was attentive) AND (individual understood the structure of the task).

For propositions listed here, operationalizations should be provided as well if possible (e.g., in experiments: participants passed attention check and comprehension questions).

Table 7.3.3: Final Consensus Auxiliary Assumptions **[only if necessary - potentially skip]** = assumptions that are secondary to the theory under investigation. They describe necessary pre-conditions to test a theory or boundary conditions. Only aspects that are not already covered should be listed here. In the column “DER-Rating,” please rate how closely each auxiliary assumption reflects the original text by entering the following number: 4 - directly taken from the text 3 - very close to the text 2 - mentioned or partially reflected in the text

1 - weakly implied by the text
0 - not mentioned in the text

In the column “Reference,” please provide the specific text snippet or passage that supports each auxiliary assumption.

Nr.	Description	Definition (Optional: Operationalization)	Reference	Comments	DER-Rating
1					
2					
…					

(7.3.4 and 7.3.5) Final Consensus Version: Problems, Insights, Further directions and Open Questions.

Please report all problems, insights, further directions and open questions in Tables 7.3.4 and 7.3.5 below.

In table 7.3.4, please indicate the following problems using these markers:
1.) A **tautology ([TAU])** occurs when the antecedent (IF) and consequence (THEN) use overlapping or identical variables, often hidden by similar wording or measurement.
→ List each in Table 7.3.4 and mark [TAU].

2.) A **contradiction ([CON])** occurs when two implications logically exclude each other (e.g., “IF frustration THEN aggression” vs. “IF frustration THEN NOT aggression”).
→ List each in Table 7.3.4 and mark [CON].

3.) An **underspecification ([USPEC])** means relations between concepts are too vague to derive clear propositions.
→ List each in Table 7.3.4 and mark [USPEC], adding a short explanation in another column.
→ Typical cases include:

- Conceptual clarity: Overlapping or poorly defined constructs.
- Timing issues: Unclear temporal sequence or dynamics of variables.

Table 7.3.4: Final Consensus Problems

= Basic requirements for any scientific theory to hold are the absence of tautologies and logical contradictions. In addition, precise definitions of all aspects of the theory should be provided.

Are there any tautologies in the specification of the theory, or is there a risk that a tautology might appear for some operationalizations of variables? Note each potential problem on a separate line and mark them with [TAU].

Are there any contradictions in the specification of the theory, or is there a risk that a contradiction might appear for some specifications? Note each potential problem on a separate line and mark them with [CON]. Are there issues in the verbal description of the theory that are underspecified to a degree that deriving implications and propositions is hard to impossible? Issues that are already solved by operationalizations and definitions are excluded here. List them, one per line, below and mark them with [USPEC]. Provide a short explanation in a separate column.			
Nr.	Description	Markers	Reference / Comments
1			
2			
…			

Table 7.3.5: Final Consensus Insights, Further directions and Open Questions Table In the process of theory specification, one often gains insights concerning i. particularly important and unexpected aspects of the theory, ii. aspects that would be important to specify further, and iii. direction for further theory development or theory integration. These insights should be listed below using one line each.			
Nr.	Description	Markers	Reference / Comments
1			
2			
…			

(Step 8; whole group) Analysis of Theory Specification Issues and Classification of Implications

As a last step, the group should fill out Table 8, which describes aspects of their final consensus version (Tables 7.3.1 to 7.3.5).

Table 8: Analysis of Theory Specification Issues and Classification of Implications

Please count and provide Answers in the format “x out of y”.

1) In how many implications did aspects have to be derived/imposed in the theory specification procedure since they were not specified by the theory? (You should have marked them with the marker “[DER]” (Table 7.3.1).

____ out of ___ implications derived/imposed

2) In how many definitions did aspects have to be derived / imposed / were indicated as missing in the theory specification procedure since they were not specified by the theory? (You should have marked them with the marker “[DER]” (Table 7.3.2.1)).

____ out of ___ definitions derived/imposed/missing

3) In how many operationalizations did aspects have to be derived / imposed / were indicated as missing in the theory specification procedure since they were not specified by the theory? (You should have marked them with the marker “[DER]” (Table 7.3.2.2).

____ out of ___ operationalizations derived/imposed/missing

Provide counts for problems detected in Table 7.3.4:

4) the number of tautologies/tautology risks detected (marker: TAU): _____

5) the number of contradictions/contradictions risks detected (marker: CON): _____

6) the number of underspecifications/underspecification risks detected (marker: USPEC): _____